

Megaloblastic — Diagnosis & Labs



1. MCV >100 opens case

WHAT YOU SEE

CASE BOARD 'MCV >100 — CASE OPENED'

WHAT IT ENCODES

Macrocytosis (MCV >100 fL; megaloblastic anemia often >110-115) triggers the workup. First branch: megaloblastic (B12/folate deficiency, antimetabolite drugs) vs non-megaloblastic (alcohol, liver disease, hypothyroidism, MDS, reticulocytosis). The presence of hypersegmented neutrophils points to megaloblastic. Mnemonic: MCV up = open the case, then split mega vs non-mega.

BOARD TRAP

WRONG: treating every high MCV as B12/folate deficiency. Discriminator: non-megaloblastic causes (alcohol, liver, hypothyroidism, MDS) have NORMAL B12/folate and NO hypersegmented neutrophils — confirm with smear + vitamin levels before replacing.

2. Hypersegmented neutrophil

WHAT YOU SEE

Neutrophil '5-6 LOBES', 'HYPERSEGMENTED'

WHAT IT ENCODES

Hypersegmented neutrophils (≥5 lobes, or any cell with 6+) are an EARLY and relatively specific clue to megaloblastic anemia and can appear before macrocytosis or anemia. They reflect impaired DNA synthesis in the granulocyte line. Mnemonic: 5+ lobes = megaloblastic flag.

BOARD TRAP

WRONG: dismissing a few hypersegmented neutrophils when the MCV is normal. Discriminator: hypersegmentation can precede anemia/macrocytosis and persists for days after treatment — it warrants B12/folate testing even with a normal CBC.

3. Oval macrocytes / anisopoikilo

WHAT YOU SEE

Red blood cell 'ANISO + POIKILO', 'OVALS'

WHAT IT ENCODES

The smear shows OVAL macrocytes (macro-ovalocytes) with anisocytosis and poikilocytosis (high RDW). Oval shape helps distinguish megaloblastic macrocytosis from the round macrocytes of liver disease/alcohol. Mnemonic: megaloblastic = oval macrocytes; non-megaloblastic = round macrocytes.

BOARD TRAP

WRONG: equating round and oval macrocytes. Discriminator: round macrocytes + target cells suggest liver disease/alcohol (non-megaloblastic); oval macrocytes + hypersegmented neutrophils suggest B12/folate deficiency (megaloblastic).

4. Hypercellular marrow, megaloblasts

WHAT YOU SEE

BONE MARROW 'HYPERCELLULAR', 'MEGALOBLASTS', 'BLOATED MEGAKARYOCYTES'

WHAT IT ENCODES

Marrow is HYPERcellular with megaloblasts (large cells with nuclear-cytoplasmic asynchrony — immature open nucleus in a mature/hemoglobinized cytoplasm) and giant bands/metamyelocytes. Despite the busy marrow, output is poor (ineffective erythropoiesis). Mnemonic: full factory, broken conveyor belt → cells die before shipping.

BOARD TRAP

WRONG: confusing megaloblastic hypercellular marrow with aplastic anemia. Discriminator: aplastic anemia is HYPOcellular (fatty marrow); megaloblastic anemia is HYPERcellular with megaloblasts and ineffective erythropoiesis (high LDH/indirect bili).

5. Mimics AML

WHAT YOU SEE

'LOOKS LIKE AML — CONFIRM FIRST'

WHAT IT ENCODES

A florid megaloblastic marrow with nuclear-cytoplasmic asynchrony can mimic AML/MDS. ALWAYS check B12, folate, MMA, and homocysteine BEFORE committing to a bone-marrow malignancy diagnosis or chemotherapy. Mnemonic: confirm the vitamins before you call it leukemia.

BOARD TRAP

WRONG (the trap): treating the marrow picture as acute leukemia immediately. Discriminator: megaloblastic change corrects fully with B12/folate replacement, whereas AML does not — a reversible vitamin deficiency must be excluded first to avoid unnecessary chemotherapy.

6. LDH & indirect bili high

WHAT YOU SEE

'LDH↑↑↑', 'UNCONJ BILI↑', 'HAPTO↓', 'URINOBILINOGEN↑'

WHAT IT ENCODES

Intramedullary (ineffective) erythropoiesis destroys precursors in the marrow → markedly HIGH LDH, high indirect (unconjugated) bilirubin, low haptoglobin, and increased urobilinogen — a hemolysis-like pattern WITHOUT external bleeding. LDH is often strikingly elevated. Mnemonic: cells die in the marrow → high LDH, high indirect bili, low hapto.

BOARD TRAP

WRONG: diagnosing primary hemolytic anemia from high LDH + indirect bili + low haptoglobin. Discriminator: in megaloblastic anemia the reticulocyte count is LOW (production failure) and the DAT/Coombs is negative, whereas true hemolysis shows a HIGH reticulocyte count.

7. Serum B12 (suspect 1)

WHAT YOU SEE

SUSPECT 1 'SERUM B12 <118-148'

WHAT IT ENCODES

A low serum B12 (roughly <148 pmol/L or <200 pg/mL; clearly deficient often <118-148) supports B12 deficiency, but the assay is imperfect — pregnancy, OCPs, and myeloma can falsely lower it, and high levels occur in liver disease/myeloproliferative states. Mnemonic: low B12 is suggestive, not definitive — there's a borderline gray zone.

BOARD TRAP

WRONG: relying on serum B12 alone in the borderline range (~150-300 ng/L). Discriminator: in the gray zone, confirm with MMA + homocysteine (both high → B12). Beware false-low B12 (pregnancy, OCP, myeloma) and false-NORMAL B12 in true deficiency (haptocorrin binding).

8. Serum folate (suspect 2)

WHAT YOU SEE

SUSPECT 2 'SERUM FOLATE'

WHAT IT ENCODES

Serum folate reflects RECENT intake and can normalize after a single good meal or rise paradoxically in B12 deficiency (methylfolate trap). A low serum folate suggests folate deficiency but is a weak standalone test. Mnemonic: serum folate = yesterday's salad, not the body's stores.

BOARD TRAP

WRONG: trusting a normal serum folate to exclude folate deficiency, or a low one to confirm it. Discriminator: use RBC folate (tissue stores) for a more reliable assessment, and remember B12 deficiency can raise serum folate via the methylfolate trap.

9. RBC folate more reliable

WHAT YOU SEE

SUSPECT 3 'RED CELL FOLATE', 'WIDER SCATTER'

WHAT IT ENCODES

RBC folate reflects tissue folate over the prior ~120 days and is more reliable than serum folate, but it FALLS in both folate AND B12 deficiency (methylfolate trap depletes usable tissue folate). Mnemonic: RBC folate = tissue truth, but it drops in both deficiencies.

BOARD TRAP

WRONG: using a low RBC folate to confirm folate deficiency specifically. Discriminator: RBC folate is low in B12 deficiency too — check B12 + MMA before labeling it isolated folate deficiency.

10. MMA high in B12 only

WHAT YOU SEE

SUSPECT 4 'MMA' yellow figure

WHAT IT ENCODES

Methylmalonic acid (MMA) is HIGH in B12 deficiency (adenosyl-B12 is the cofactor for methylmalonyl-CoA mutase) but NORMAL in folate deficiency — the single most useful test to separate the two and to resolve a borderline B12. Mnemonic: MMA = B12's fingerprint (folate leaves no print).

BOARD TRAP

WRONG (key distractor): 'MMA is high in both B12 and folate deficiency.' It is NOT — folate has a NORMAL MMA. Discriminator: high MMA + high homocysteine = B12; normal MMA + high homocysteine = folate.

11. Homocysteine high in both

WHAT YOU SEE

SUSPECT 5 'HOMOCYSTEINE'

WHAT IT ENCODES

Homocysteine is elevated in BOTH B12 and folate deficiency (both are methionine-synthase cofactors), so it confirms a deficiency exists but cannot tell which vitamin. Pair it with MMA to separate them. Mnemonic: homocysteine = shared road; MMA = the fork.

BOARD TRAP

WRONG: using homocysteine to distinguish B12 from folate. Discriminator: it rises in both — only MMA (high in B12, normal in folate) makes the distinction. Isolated high homocysteine with normal MMA favors folate (or B6/MTHFR causes).

12. Borderline → MMA + homocysteine

WHAT YOU SEE

'BORDERLINE ZONE', '>100-200 ng/L'

WHAT IT ENCODES

For an equivocal/borderline B12 (~150-300 ng/L), do NOT decide on the level alone — order MMA + homocysteine. BOTH elevated → true B12 deficiency; both normal → B12 deficiency effectively excluded. Mnemonic: gray-zone B12? Let MMA + homocysteine be the tiebreaker.

BOARD TRAP

WRONG: calling a borderline B12 'normal' and stopping. Discriminator: functional deficiency hides in the gray zone — confirm with metabolites (MMA/homocysteine) and treat if elevated, especially when neuro signs are present.

13. Iron deficiency masks macro

WHAT YOU SEE

'Fe MASKS', 'MCV OK — CHECK B12', RDW high

WHAT IT ENCODES

Coexisting iron deficiency (microcytic) offsets the macrocytosis of B12/folate deficiency → a NORMAL MCV but HIGH RDW and a dimorphic smear (both micro- and macrocytes). Mnemonic: micro + macro = normal-looking MCV; the high RDW betrays it.

BOARD TRAP

WRONG (trap): 'a normal MCV rules out megaloblastic anemia.' It does NOT. Discriminator: a high RDW, hypersegmented neutrophils, or anemia with a normal MCV should prompt iron studies + B12/folate — combined deficiencies hide behind a normal MCV.

14. Reticulocyte count low

WHAT YOU SEE

Top-left detective with a magnifying glass collecting blood-cell 'evidence' at the crime scene — the investigator gathering clues represents the diagnostic work-up that uncovers an inappropriately LOW reticulocyte count (production failure, not hemolysis).

WHAT IT ENCODES

Despite the hemolysis-like labs (high LDH/indirect bili), the reticulocyte count is INAPPROPRIATELY LOW because the underlying problem is production failure (ineffective erythropoiesis), not increased peripheral destruction. After treatment, a reticulocyte response peaks at days 5-7. Mnemonic: marrow can't ship cells → low retics now, retic spike after vitamins.

BOARD TRAP

WRONG: a high LDH + low haptoglobin leading to a primary hemolysis diagnosis. Discriminator: a LOW reticulocyte count plus oval macrocytes and hypersegmented neutrophils points to megaloblastic production failure, not hemolysis (which would show high reticulocytes).